

ISC Class XII Maths — Chapter 5

Continuity & Differentiability — Formula Sheet

A. Continuity

f is continuous at $x = c \iff \lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = f(c)$, i.e. LHL = RHL = $f(c)$.

$$\text{LHL} = \lim_{h \rightarrow 0} f(c - h) \quad \text{RHL} = \lim_{h \rightarrow 0} f(c + h)$$

Algebra of continuous functions: if f, g are continuous at c , so are $f \pm g$, fg , kf , and f/g (where $g(c) \neq 0$); the composite $f(g(x))$ is continuous wherever g is continuous at c and f is continuous at $g(c)$.

$$|x - a| = \begin{cases} x - a, & x \geq a \\ -(x - a), & x < a \end{cases} \text{ — split before taking one-sided limits.}$$

B. Differentiability

$$Lf'(c) = \lim_{h \rightarrow 0^-} \frac{f(c + h) - f(c)}{h}$$

$$Rf'(c) = \lim_{h \rightarrow 0^+} \frac{f(c + h) - f(c)}{h}$$

f is differentiable at $c \iff Lf'(c) = Rf'(c)$ (both finite and equal).

Differentiable $\Rightarrow$ continuous, but **not conversely** — e.g. $f(x) = |x|$ is continuous but not differentiable at 0.

C. Standard Trigonometric Limits

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

$$\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1$$

$$\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} = 0$$

$$\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2} = \frac{1}{2}$$

Tip: Always manipulate the given limit algebraically until it matches $\frac{\sin(\text{something} \rightarrow 0)}{\text{that same something}}$ before applying the standard result — multiply/divide to create matching angles, as in $\lim \frac{x}{\sin 3x} = \lim \frac{\frac{1}{3}}{\frac{1}{3}} \cdot \frac{3x}{\sin 3x}$.

D1. Pythagorean & Sum/Diff

Pythagorean:

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

Sum/Difference:

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

D2. Double & Half Angle

Double angle:

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Half-angle (for simplification):

$$1 - \cos x = 2 \sin^2 \frac{x}{2} \quad 1 + \cos x = 2 \cos^2 \frac{x}{2}$$

$$1 \pm \sin x = \left(\cos \frac{x}{2} \pm \sin \frac{x}{2} \right)^2$$

D3. Sum-to-Product

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

D4. Co-function Identities

Using $\left(\frac{\pi}{2} - x\right)$:

$$\sin x = \cos\left(\frac{\pi}{2} - x\right)$$

$$\tan x = \cot\left(\frac{\pi}{2} - x\right)$$

$$\sec x = \csc\left(\frac{\pi}{2} - x\right)$$

E1. Derivatives — Algebraic, Exp, Log

$$\frac{d}{dx}(x^n) = nx^{n-1} \quad \frac{d}{dx}(k) = 0$$

$$\frac{d}{dx}(e^x) = e^x \quad \frac{d}{dx}(a^x) = a^x \ln a$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x} \quad \frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$$

E2. Derivatives — Trigonometric

$$\frac{d}{dx} \sin x = \cos x \quad \frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \tan x = \sec^2 x \quad \frac{d}{dx} \cot x = -\csc^2 x$$

$$\frac{d}{dx} \sec x = \sec x \tan x \quad \frac{d}{dx} \csc x = -\csc x \cot x$$

E3. Derivatives — Inverse Trig

$$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}} \quad \frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2} \quad \frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$$

$$\frac{d}{dx} \sec^{-1} x = \frac{1}{|x|\sqrt{x^2-1}} \quad \frac{d}{dx} \csc^{-1} x = \frac{-1}{|x|\sqrt{x^2-1}}$$

F. Chain, Product & Quotient Rules

Chain rule: if $y = f(u)$, $u = g(x)$: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

Product rule: $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$

Quotient rule: $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

Worked Pattern: Inverse-Trig via Substitution

Before differentiating $\sin^{-1}/\cos^{-1}/\tan^{-1}$ of a messy expression, substitute to collapse it to a single angle:

$$x = \sin \theta \Rightarrow \sqrt{1-x^2} = \cos \theta$$

$$x = \tan \theta \Rightarrow \sqrt{1+x^2} = \sec \theta$$

$$x = \cos \theta \Rightarrow \sqrt{1-x^2} = \sin \theta$$

For $\sqrt{\frac{1-\cos x}{1+\cos x}}$ -type: use half-angle (Sec. D2) to reduce to $\tan \frac{x}{2}$, so $\tan^{-1}(\cdot) = \frac{x}{2}$, giving $\frac{dy}{dx} = \frac{1}{2}$ instantly.

G. Implicit Differentiation

Differentiate *both sides* w.r.t. x , treating y as a function of x : every term in y picks up a factor $\frac{dy}{dx}$ (chain rule), e.g. $\frac{d}{dx}(y^2) = 2y \frac{dy}{dx}$.

Then collect all $\frac{dy}{dx}$ terms on one side and solve.

Worked Pattern: Logarithmic Differentiation

Use when the function is a **product/quotient of many factors**, or has a **variable in both base and exponent** (u^v):

1. Take $\ln$ of both sides: $\ln y = \ln(\text{RHS})$.
2. Expand using $\ln(ab) = \ln a + \ln b$, $\ln(a/b) = \ln a - \ln b$, $\ln(a^n) = n \ln a$.
3. Differentiate implicitly: LHS gives $\frac{1}{y} \frac{dy}{dx}$.
4. Multiply through by y (the original function) to isolate $\frac{dy}{dx}$.

H. Parametric Differentiation

If $x = f(t)$, $y = g(t)$: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

Second derivative: $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}$ — differentiate $\frac{dy}{dx}$ w.r.t. t again, then divide by dx/dt (never divide by d^2x/dt^2).

I. Function w.r.t. Another Function

To find $\frac{dy}{du}$ where both y, u are functions of x :
 $\frac{dy}{du} = \frac{dy/dx}{du/dx}$

Reciprocal rule: $\frac{dy}{dx} = \frac{1}{dx/dy}$ — useful when x is easier to isolate than y (isolate x in terms of y , differentiate, then reciprocate).

Worked Pattern: Infinite Recursive Forms

For $y = \sqrt{f(x) + \sqrt{f(x) + \sqrt{f(x) + \cdots}} (\infty)}$:

1. Use self-similarity: the expression under the first root is the whole thing again, so $y = \sqrt{f(x) + y}$.
2. Square both sides: $y^2 = f(x) + y$.
3. Differentiate implicitly and solve for $\frac{dy}{dx}$.

Same idea for infinite continued fractions/powers — rewrite using self-similarity first, always *before* differentiating.

J. Second-Order Derivatives

$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$ — differentiate $\frac{dy}{dx}$ again w.r.t. x (product/chain rule as needed).

For implicit relations: find $\frac{dy}{dx}$ first (implicitly), then differentiate that expression again implicitly — substitute the first-derivative result back in to simplify $\frac{d^2y}{dx^2}$ fully in terms of x, y only.

K. Common Traps to Avoid

- Continuity needs **all three** equal: LHL, RHL, and $f(c)$ — matching limits alone isn't enough if $f(c)$ differs.
- Differentiability $\Rightarrow$ continuity, but the converse is false — $|x|$ is the classic counterexample.
- In logarithmic differentiation, don't forget to multiply back by the *original* y at the end.
- Parametric $\frac{d^2y}{dx^2} \neq \frac{d^2y/dt^2}{d^2x/dt^2}$ — must differentiate dy/dx w.r.t. t , then divide by dx/dt .
- Every term containing y needs a $\frac{dy}{dx}$ factor in implicit differentiation — easy to drop on product terms like xy (needs product rule: $x\frac{dy}{dx} + y$).

L. Quick Recognition Guide

Pattern seen	Use
Piecewise function, “examine continuity at $x = c$ ”	Sec. A — LHL, RHL, $f(c)$
$\frac{0}{0}$ form with $\sin/\tan$ as $x \rightarrow 0$, find k	Sec. C standard trig limits
$ x-a $ inside the function	Sec. A — split into cases first
“Is f derivable/differentiable at...”	Sec. B, $Lf' = Rf'$
$\tan^{-1}/\cos^{-1}$ of a messy fraction before “differentiate”	Worked Pattern: Inverse-Trig Substitution
$\sqrt{\frac{1 - \cos x}{1 + \cos x}}$ or similar inside inverse trig	Sec. D2 half-angle first
Product of 3+ factors, or u^v form	Worked Pattern: Logarithmic Differentiation
$x = f(t)$, $y = g(t)$ given	Sec. H parametric
Equation mixing x, y with no easy $y = \dots$ form	Sec. G implicit
“Find $\frac{dy}{du}$ ” with both in terms of x	Sec. I
Infinite root/fraction “ $\dots \infty$ ”	Worked Pattern: Infinite Recursive Forms
Asked for $\frac{d^2y}{dx^2}$	Sec. J (direct or implicit)